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Assignment - 1

V/w ✓

Evolution of Database:

The evolution of database system refers to the development of methods for storing, organizing and managing data efficiently. Over time, database systems have evolved through several stages:

File - Based systems (1950s - 1960s)

- * Data was stored in separate files.
- * Each application maintained its own files.
- * Problems included data redundancy, inconsistency, and difficult data sharing.

Hierarchical Database Model (1960s)

- * Data was organized in a tree-like structure with parent-child relationships.
- * Fast for one-to-many relationships.
- * Example: IBM IMS.
- * Limitation: Rigid structure and

difficult to modify.

Network Database Model (1970s)

- * Data was organized as a graph allowing many-to-many relationships.

- * More flexible than the hierarchical model.

- * Limitation: Complex database design and navigation.

Relational Database Model (1970s - Present)

- * Introduced by E.F. Codd.

- * Data is stored in tables (rows and columns).

- * Uses SQL (Structured Query Language) for data manipulation.

- * Examples: MySQL, Oracle, SQL Server,

PostgreSQL,

- * Advantages: Reduced redundancy, easy querying and high data integrity

Object - Oriented Database Systems (1990s)

- * stores data as objects similar to object oriented programming.
- * Suitable for multimedia, engineering, and Scientific application.

NoSQL and Cloud Database (2000s - Present)

- * Designed for handling large volume of unstructured and semi structured data.
- * Supports high scalability and distributed storage.
- * Example: MongoDB, Cassandra, Fier base.
- * Widely used in big data, cloud computing and Web application.

Conclusion:

* Database systems have evolved from simple, file based storage to advance relational, NOSQL and cloud database. making them essential for today's application.